

1. Introduction & Literature Review

Cyber-attacks are becoming more common, and many traditional security systems cannot detect new attacks very well. Machine learning may work better because it can learn patterns in network traffic and decide if the traffic is benign or malicious.

DDoS attacks are especially dangerous because they can overload a network and stop people from using a website or service. Researchers in Machine Learning-Based Intrusion Detection Systems found that Random Forest performed very well for detecting malicious traffic. The AI for Threat Detection also explains that machine learning can help detect attacks faster.

Companies such as Microsoft, IBM, and Google already use machine learning in their security tools. However, many datasets are imbalanced and may not fully represent real network traffic, so this project compares several models to see which one works best.

2. Problem Definition

The goal of this project is to use machine learning to classify network traffic as either benign or malicious. This project focuses on detecting DNS-based DDoS attacks.

The input features include information from the network traffic, such as protocol type, flow duration, packet counts, and packet sizes. The output is a binary label where 0 means benign traffic and 1 means malicious traffic.

This is a supervised machine learning classification problem because the dataset already includes the correct labels for each network traffic record.

3. Approach, Materials, and Methods

This project used the CICDDoS2019 dataset, which contains both benign and malicious network traffic. Because the full dataset was too large, a subset of 50,000 records was used in Google Colab.

The dataset was cleaned before training the models. The Label column was changed so that benign traffic = 0 and malicious traffic = 1. Text columns such as Flow ID, Source IP, Destination IP, Timestamp, and SimilarHTTP were removed because the models only work with numbers. Missing values and infinite values were also fixed.

The data was split into 80% training data and 20% testing data. Four machine learning models were tested:

- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbors (KNN)

Logistic Regression was used as a simple starting model. Decision Tree and Random Forest were chosen because they are often used in cybersecurity research, and KNN was added to compare a different type of model.

The project was completed in Python using Google Colab with the libraries pandas, numpy, scikit-learn, and matplotlib.

4. Results

Four machine learning models were tested and compared using accuracy.

Model	Accuracy
Random Forest	99.99%
Decision Tree	99.98%
K-Nearest Neighbors	99.71%
Logistic Regression	98.80%

Random Forest had the highest accuracy and was chosen as the final model. Decision Tree also performed very well, while Logistic Regression had the lowest accuracy.

Random Forest had perfect precision, recall, and F1-score for both benign and malicious traffic.

Metric	Benign (0)	Malicious (1)
Precision	1.00	1.00
Recall	1.00	1.00
F1-Score	1.00	1.00

Logistic Regression performed worse on benign traffic because the dataset had many more malicious records than benign records.

Metric	Benign (0)	Malicious (1)
Precision	0.81	0.99
Recall	0.82	0.99
F1-Score	0.82	0.99

The results improved as more advanced models were tested. Logistic Regression was the starting model, and Random Forest gave the best results.

5. Discussion

All four models worked well, but Random Forest gave the best results with 99.99% accuracy.

Random Forest worked best because it combines many decision trees, which helps it find patterns in the network traffic better than the other models.

Logistic Regression had the lowest accuracy because it had more trouble identifying benign traffic. The dataset had many more malicious records than benign records, which made the data imbalanced.

One challenge in this project was that the original dataset was too large for Google Colab, so only 50,000 records were used. Another limitation is that the dataset only included one type of attack, so the results may be different with other types of network traffic.

These results are like what researchers and companies such as Microsoft, IBM, and Google have found. Machine learning, especially Random Forest, can be very useful for detecting malicious network traffic.

6. Conclusion

This project showed that machine learning can be used to classify network traffic as benign or malicious. Four models were tested, and Random Forest gave the best results.

I also learned that cleaning and preparing the dataset is very important before training the models. In the future, this project could be improved by using more attack types and a more balanced dataset.